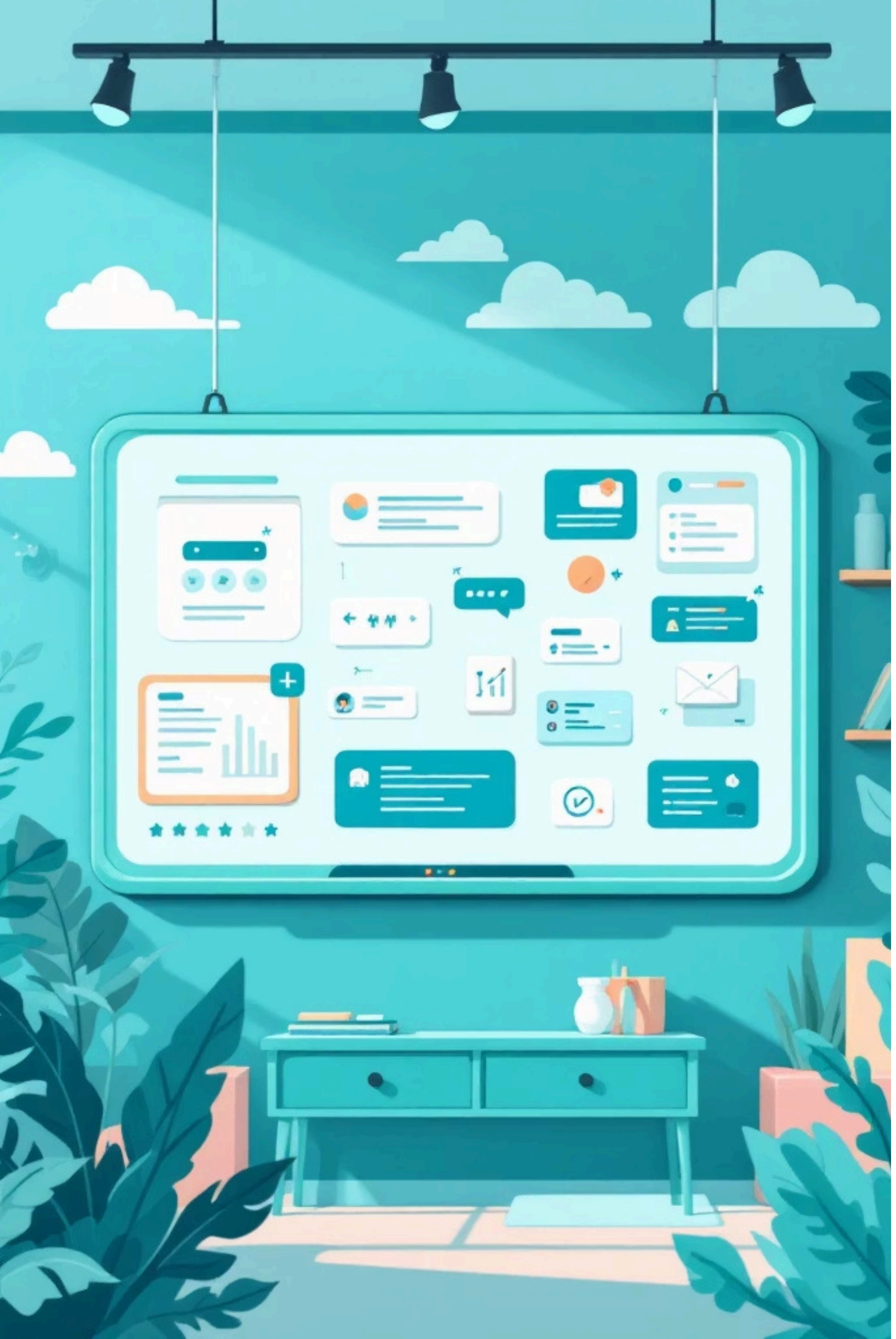


Muralia

A Collaborative Web Platform for Education

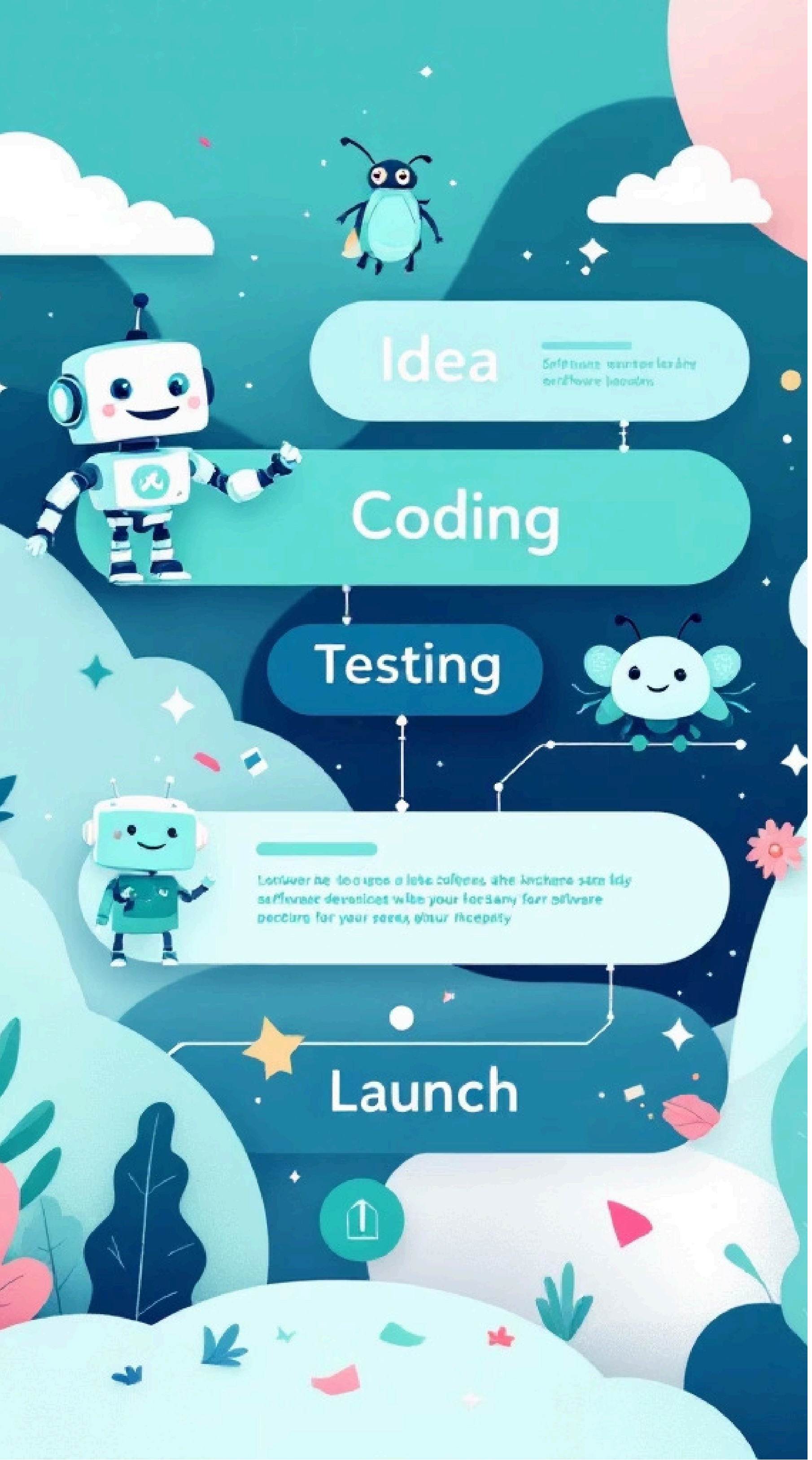
An intermodular development project created for Campus Cámara Sevilla's Digital Applications Development programme, designed to empower educators and students with unlimited collaborative workspace capabilities.

Project Purpose

Unlimited Collaboration

Muralia provides a comprehensive digital whiteboard platform that enables real-time collaboration amongst students and educators. Inspired by industry standards such as Padlet and Miro, this solution eliminates restrictive usage limits whilst maintaining robust functionality.

- Unlimited boards for diverse projects
- Real-time collaboration across multiple users
- Rich multimedia notes and content embedding
- Intuitive interface for seamless workflow



Development Methodology

Agile Scrum Framework

Iterative development using two-week sprints with daily stand-ups, sprint planning, and retrospectives to ensure continuous improvement and stakeholder alignment.

Angular Frontend

Modern component-based architecture implementing the MVVM pattern for efficient data binding and maintainable code structure.

Node.js Backend

Express framework with MVC architecture providing RESTful API endpoints for seamless frontend-backend communication.

Database Architecture

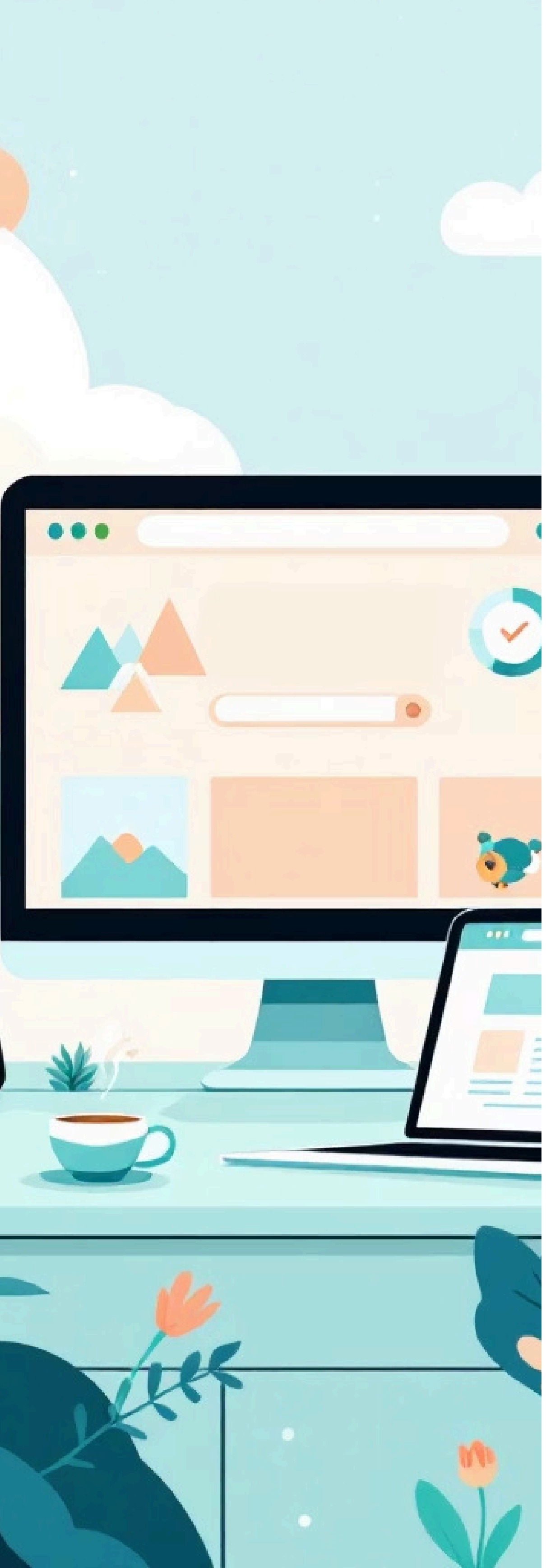
Strategic Choice: MongoDB

NoSQL document-based database selected for optimal flexibility with dynamic, unstructured data requirements.

Why MongoDB?

The unstructured nature of collaborative whiteboard data—with variable note formats, multimedia attachments, and dynamic user interactions—demands flexible schema design. MongoDB's document-oriented structure provides:

- Schema flexibility for evolving requirements
- Efficient handling of diverse data types
- Horizontal scalability for growing user base
- JSON-like documents for intuitive data mapping



Technical Challenges & Solutions

Real-Time Synchronisation

Implemented WebSocket connections to ensure instant updates across all connected users. Changes propagate immediately without page refresh, maintaining collaborative flow.

Responsive Design

CSS Grid and Flexbox architecture adapts seamlessly from mobile devices to desktop monitors, ensuring consistent experience across all screen sizes and orientations.

Security Implementation

JWT-based authentication with encrypted tokens provides secure user sessions whilst preventing session hijacking and cross-site request forgery attacks.

Accessibility Standards

WCAG 2.1 compliance ensures platform accessibility for users with disabilities through semantic HTML, ARIA labels, keyboard navigation, and contrast optimisation.

Key Learnings & Impact

Technical Mastery

This project demonstrated comprehensive full-stack development skills, integrating frontend Angular components with Node.js backend services whilst implementing secure authentication and real-time communication protocols.

Beyond Coding

The development process emphasised understanding user needs, conducting requirements analysis, and delivering practical solutions that address real educational challenges rather than simply implementing features.

Project Success

Muralia successfully delivers an unrestricted collaborative platform that empowers educators and students with flexible digital workspace capabilities, bridging the gap between traditional classroom methods and modern web technologies.